

Resource Allocation for URLLC/mMTC in 6G Industrial IoT Networks: Joint Optimization under Mixed-Numerology and SNR Constraints

Zhuofan Cui, Heba Dawoud, Faleh Alshalwi, Yusuf Sambo, Muhammad Ali Imran, and Olaoluwa Popoola
James Watt School of Engineering, University of Glasgow, Glasgow, UK
Email: {2969953Z, 2810361D, 2958470A} @student.gla.ac.uk,
and {Yusuf.Sambo, Muhammad.Imran, Olaoluwa.Popoola}@glasgow.ac.uk

Abstract—Modern smart factories accommodate numerous heterogeneous devices, making co-scheduling of Ultra-Reliable Low-Latency Communications (URLLC) and massive Machine-Type Communications (mMTC) services on shared physical resources a critical challenge. This paper proposes a resource allocation mechanism for this convergence scenario, combining Signal-to-Noise Ratio (SNR) boosting strategy, flexible numerology configuration, and device priority control. By formulating a multi-objective optimization model that integrates delay constraints, channel conditions, and resource utilization, we design a low-complexity adaptive scheduling algorithm. The simulation results demonstrate that our proposed method improves URLLC schedulability by 19.8% while achieving an average latency of 0.843 ms, substantially outperforming the baseline algorithms. This paper provides effective technical support for the establishment of industrial wireless communication systems capable of supporting coexisting Quality of Service (QoS) requirements.

Index Terms—URLLC, mMTC, flexible numerology, resource allocation, heuristic QoS-aware scheduling.

I. INTRODUCTION

The paradigm shift toward Industry 4.0 has accelerated the evolution of smart factories from traditional wired infrastructures to advanced wireless communication systems, enabling unprecedented flexibility and scalability in industrial automation. This transformation requires the deployment of heterogeneous wireless devices with different QoS requirements within shared spectral resources, where environmental sensors align with mMTC characteristics due to tolerance of delays, while mission-critical applications such as robotic control systems demand URLLC stringent temporal and reliability constraints [1], [2]. The forthcoming Sixth-Generation (6G) wireless systems, as envisioned by the International Telecommunication Union (ITU) International Mobile Telecommunications (IMT)-2030 framework, promise sub-0.1 ms latency, 10^{-7} reliability, and ultra-high device densities [3], necessitating fundamental innovations in resource allocation mechanisms for diverse service co-existence while maintaining stringent QoS guarantees [4].

Recent advances in URLLC resource allocation for industrial environments have shown promising results. Li et al. [5] proposed a Fifth-Generation (5G) New Radio (NR)-compliant framework using matching game theory and water filling algorithms for joint time-frequency and power allocation,

while Sharara et al. [6] developed an Open-RAN (ORAN) architecture integrating transformer networks with Mixed-Integer Linear Programming (MILP) for dynamic scheduling, highlighting adaptive numerology benefits. For mMTC applications, Sadi et al. [7] introduced semi-persistent scheduling with multi-numerology configurations to enhance spectral reuse efficiency, and flexible multi-numerology systems have emerged as key enablers for various 6G industrial applications [8], [9].

However, these contributions reveal critical limitations when examined in real-world industrial contexts. First, existing flexible numerology approaches assume idealized channel conditions, overlooking complex industrial propagation characteristics [6], [7]. Second, contemporary URLLC methods employ fixed numerology configurations, lacking adaptive flexibility for varying QoS demands [5]. Third, the current literature addresses the services URLLC and mMTC services in isolation, providing insufficient solutions for service co-existence and resource contention within shared pools [5], [7], [9]. These gaps significantly limit the effectiveness of existing approaches in supporting the heterogeneous convergence of services required for 6G smart factory environments.

This paper addresses these challenges by proposing a QoS-aware resource allocation framework for heterogeneous 6G industrial networks that jointly optimizes URLLC and mMTC services through integrated channel-aware SNR enhancement, flexible numerology adaptation, and multi-objective optimization to achieve significant latency improvements while guaranteeing stringent QoS requirements.

The main contributions of this work can be summarized as follows:

- 1) We develop a novel SNR boosting strategy that selectively enhances transmit power for critical URLLC devices under Non-Line-of-Sight (NLoS) conditions, improving schedulability and ensuring reliability in quasi-static channel environments.
- 2) We formulate a multi-objective optimization problem that minimizes bandwidth utilization and communication latency for joint URLLC-mMTC resource allocation under practical constraints.

- 3) We design a low-complexity adaptive scheduling algorithm integrating SNR-based filtering, numerology mapping, and priority-aware allocation, enabling sub-millisecond latency for URLLC.
- 4) The simulation results demonstrate that the proposed algorithm improves URLLC scheduling by 19.8% and achieves an average latency of 0.843 ms, outperforming the baseline algorithms.

The remainder of this paper is organized as follows. Section II establishes the system model and the characterization of the industrial channel. Section III presents the problem formulation and optimization framework. Section IV details the adaptive scheduling algorithm. Section V provides performance evaluation and comparative analysis. Section VI concludes the paper.

II. SYSTEM MODEL

A. System Architecture

We consider a typical Industrial Internet of Things (IIoT) scenario with a single 5G base station gNodeB (gNB) managing resource allocation for heterogeneous devices in a factory environment [10]. As illustrated in Fig. 1, gNB is positioned 10 meters outside the factory to ensure full coverage.

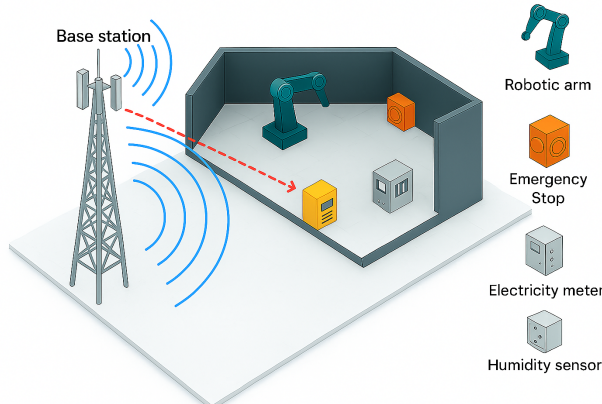


Fig. 1. Scenario and System Model.

A total of $I = 200$ devices, denoted by the set $\{i_1, i_2, \dots, i_N\}$, are randomly distributed and equipped with single antennas at height $h_d = 1.5$ m. Devices connect to the base station for downlink control and data, while uplink and inter-cell interference are deliberately excluded owing to the factory's isolated operational setting. This heterogeneous device population comprises two distinct categories: mission-critical URLLC devices and massive mMTC devices.

B. Channel Model

This work considers a channel model for communication between static industrial devices in suburban factory settings, following the Indoor Factory with Sparse clutter and Low base station height (InF-SL) scenario defined in 3GPP TR 38.901 with center frequency $f_c = 3.5$ GHz [11], [12]. Given the

stationary nature of both the equipment and the base station infrastructure, we adopt a quasi-static channel model [5]. This model neglects the effects of small-scale fast fading and considers only large-scale fading, which encompass Path Loss (PL) and lognormal shadow fading. This approach eliminates the necessity to consider real-time Channel Quality Indicator (CQI) and Channel State Information (CSI) dynamics.

1) **Line-of-Sight (LoS) Probability Model:** In this context, 3GPP defines a distance-dependent LoS probability model. Let d_{2D} represent the distance between the base station gNB and the device i . As described in [11], the probability of the device being in the LoS state at this distance can be expressed in an exponentially decaying form, as shown in the following equation:

$$\Pr_{\text{LoS,subsec}}(d_{2D}) = \exp\left(-\frac{d_{2D}}{k_{\text{subsec}}}\right), \quad (1)$$

the parameter $k_{\text{subsec}} = \frac{-d_{\text{clutter}}}{\ln(1-r)}$ is the environmental attenuation distance, which is determined by the factory layout, derived from the average obstacle spacing d_{clutter} and the empirical value r .

2) **Propagation Distance Modeling:** We decompose the total propagation distance into two segments: the outdoor and indoor parts, given the existence of a propagation path from the outdoor to the indoor. The total 3D propagation distance $d_{3D,\text{total}}$ is calculated in [11] as:

$$d_{3D,\text{total}} = \sqrt{(d_{2D,\text{out}} + d_{2D,\text{in}})^2 + (h_{\text{BS}} - h_{\text{UT}})^2}, \quad (2)$$

where $d_{2D,\text{out}}$ and $d_{2D,\text{in}}$ are the horizontal distances from the base station to the exterior wall of the building and the horizontal distances of signal propagation in the interior, respectively. h_{BS} and h_{UT} are the heights of the base station and antenna, respectively. The formula combines outdoor and indoor transmission paths into a single 3D propagation distance for PL calculation, which is discussed in the next subsection.

3) **PL Models:** For the LoS condition, the PL is usually modelled as the logarithm of distance and center frequency. Therefore, the LoS PL model for the InF-SL environment at center frequency f_c (GHz), presented in [11] can be expressed as follows:

$$PL_{\text{LoS}} = 31.84 + 21.50 \log_{10}(d_{3D}) + 19.00 \log_{10}(f_c). \quad (3)$$

However, in the NLoS scenario, there is a high density of clutter within the plant, and much of the equipment is obscured by objects such as shelves and walls. Therefore, the NLoS PL is modelled as [11]:

$$PL_{\text{NLoS}}^{\text{InF-SL}} = 33 + 25.5 \log_{10}(d_{3D}) + 20 \log_{10}(f_c), \quad (4)$$

$$PL_{\text{NLoS}} = \max(PL, PL_{\text{LoS}}). \quad (5)$$

The NLoS scenario exhibits a higher distance attenuation factor, indicating a significant added attenuation in densely cluttered settings.

4) **SNR Calculation:** Using the aforementioned large-scale fading channel model, we modelled the SNR of each device through the discriminative mechanism $\text{Pr}_{\text{LoS,subsc}}$. The SNR of a single channel can be determined from the transceiver parameters and PL. The base station transmitter power is denoted as P_{tx} , the transmit and receive antenna gains are indicated as G_{tx} and G_{rx} , respectively, while the PL after inclusion of shadowing fading is represented as PL . The receiver noise power is N_0 , which can be calculated from the Boltzmann constant. Therefore, the SNR is given by:

$$\text{SNR}_i = P_{\text{tx}} + G_{\text{tx}} + G_{\text{rx}} - PL - N_0. \quad (6)$$

The primary role of the SNR in this work is to measure each device's wireless channel, acting as a crucial factor in deciding schedulability.

C. Flexible Time-frequency resource structure in 5G NR

To accommodate diverse QoS needs, especially for URLLC in industrial settings, all devices and the base station utilize a full bandwidth of $B_{\text{tot}} = 100$ MHz within a licensed industrial spectrum [12].

5G NR introduces a flexible numerology framework, where each numerology corresponds to a specific Subcarrier Spacing (SCS) and symbol duration [7]. This paper considers three numerology configurations; see Table I [7], [13], with subcarrier spacings $\Delta f = 15, 30$, and 60 kHz, allowing adaptable transmission intervals from milliseconds to sub-milliseconds, well-suited for heterogeneous traffic. To meet the stringent latency demands of URLLC, a two-symbol mini-slot structure is adopted for $\Delta f = 60$ kHz, with a duration of $16.67 \mu\text{s}$, capable of completing downlink transmissions within the 1 ms latency constraint [14]. In addition, we formulate the resource allocation framework by defining a Resource Block (RB) as the fundamental scheduling unit, representing a fixed number of subcarriers across the frequency domain and varying mini-slot or slot durations determined by time-domain numerology.

III. PROBLEM FORMULATION

We tackle the cooperative scheduling challenges of URLLC and mMTC devices with limited wireless resources by creating a multi-objective optimization framework. Extending the approach in [7], this joint optimization minimizes total bandwidth usage and incorporates the average communication latency per device, ensuring that the QoS requirements of each device are met.

To balance the trade-off between bandwidth efficiency and latency performance, we formulate a weighted multi-objective optimization. Let $\alpha \in [0, 1]$ and $\beta = 1 - \alpha$ represent weighting factors to optimize resource utilization and latency, respectively. Both latency and bandwidth usage are normalized to ensure fairness and prevent one objective from dominating the other. The overall problem is mathematically defined as an optimization problem $\mathcal{P}$, as described below:

$$\min \quad \alpha \sum_{n=1}^{n_{\max}} z_n f_n + \beta L_{\text{avg}} \quad (7)$$

$$\text{s.t.} \quad \sum_{n=1}^{n_{\max}} x_{in} = 1, \quad \forall i \in \mathcal{I} \quad (7a)$$

$$\sum_{i \in \mathcal{I}} x_{in} \leq |\mathcal{I}| z_n, \quad \forall n \in [1, n_{\max}] \quad (7b)$$

$$\tau_n + \sum_{i \in \mathcal{H}_i} [p_i/p_1] \tau_n \leq L_i, \quad \forall i \in \mathcal{I}, \forall n \in [1, n_{\max}] \quad (7c)$$

$$\text{SNR}_i \geq \text{SNR}_{\text{threshold}}, \quad \forall i \in \mathcal{I}_{\text{candidate}}. \quad (7d)$$

The above objective function minimizes total bandwidth usage and average latency across the scheduled device set $\mathcal{I}$. The bandwidth usage is measured by the number of RBs allocated, where each device is assigned one RB from a set of bands $n \in \{1, \dots, n_{\max}\}$, each linked to a numerology type $\mu \in \{0, 1, 2\}$. A binary variable z_n indicates whether any mMTC device uses the band n , and f_n represents the bandwidth per resource block for selecting the band, calculated by $f_n = 12 \times \Delta f$ [7]. The second component, L_{avg} , captures the average latency (ms) between scheduled devices, incorporating both transmission and queueing delays. This term is optimized to meet the stringent QoS requirements of URLLC devices, which can be defined as:

$$L_{\text{avg}} = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \sum_{n=1}^{n_{\max}} x_{in} \cdot \left(\tau_n + \sum_{l \in \mathcal{H}_i} \left[\frac{p_i}{p_l} \right] \tau_n \right), \quad (8)$$

where $|\mathcal{I}|$ denotes the total number of devices that have successfully received resource allocation in device $\mathcal{I}$. Each device $i \in \mathcal{I}$ is characterized by a fixed packet generation period p_i . The set $\mathcal{H}_i$ denotes devices with a higher scheduling priority than the device i , and τ_n represents the transmission latency for data transmission. p_l denotes the packet generation period of $\mathcal{H}_i$, which is used to calculate the queueing delay.

The optimization problem includes several practical constraints to ensure a feasible allocation of resources. Constraint (7a) enforces exclusive band assignment, where the binary variable $x_{in} = 1$ indicates that device i occupies the band n , preventing re-assignment. Constraint (7b) mandates band activation when devices are assigned. Constraint (7c) limits total latency L_{total} , comprising queuing and transmission delays, to not exceed the QoS-specific maximum latency requirement L_i for each device i . Constraint (7d) ensures channel reliability by requiring all scheduled devices to maintain the SNR above the predetermined threshold $\text{SNR}_{\text{threshold}}$.

This problem is classified as Mixed-Integer Nonlinear Programming (MINLP) due to discrete binary variables (z_n and x_{in}) coupled with nonlinear latency calculations. Being NP-hard, MINLP problems lack polynomial-time optimal solutions. Therefore, we demonstrate the effectiveness of the proposed algorithm by verifying that the latency and bandwidth requirements satisfy the QoS constraints.

TABLE I
5G NR NUMEROLOGY AND APPLICABLE SCENARIOS

Numerology (μ)	SCS (kHz)	Slot Duration (ms)	Cyclic Prefix	Symbol Duration (μ s)	Applicable Scenarios
0	15	1.0	Normal	≈ 66.67	eMBB, mMTC
1	30	0.5	Normal	≈ 33.33	mMTC, moderate URLLC
2	60	0.25	Normal	≈ 16.67	URLLC, low latency

IV. PROPOSED SOLUTION

The proposed algorithm addresses the NP-hard problem of binary scheduling variables and nonlinear delay constraints by integrating service prioritization and channel quality-aware resource allocation. Unlike existing work [7], our approach introduces a novel SNR enhancement mechanism and multi-numerology mapping strategy for low-latency URLLC devices, while maintaining fairness for mMTC devices.

A. Initialization Phase

During the initialization phase, devices are classified and wireless resources are prepared for subsequent SNR estimation and resource allocation. Devices are then prioritized according to the service type: URLLC devices are assigned Priority = 1, while mMTC are designated with Priority = 0, which directly influences their scheduling order. Subsequently, the channel conditions are assessed by estimating the LoS probability $\text{Pr}_{\text{LoS,subsc}}$ for each device. The resulting value is stored as $\text{deviceLOS}[i]$, guiding subsequent SNR enhancement strategies and scheduling decisions.

B. SNR Estimation and Boosting Phase

The purpose of this stage is to perform an accurate assessment of device wireless channel quality prior to resource allocation, thereby enhancing resource utilization and ensuring communication reliability. We begin by calculating the SNR of each device in sets $\text{DeviceList}_{\text{URLLC}}$ and $\text{DeviceList}_{\text{mMTC}}$ using the initial transmit power $P_{\text{tx_global}}$, subsequently filtering those that exceed the threshold $\text{SNR}_{\text{threshold}}$. To improve the reliability of mission-critical communications, a transmit power boosting strategy is introduced. Specifically, URLLC devices experiencing NLoS conditions with sub-threshold SNR values are virtually re-evaluated under an elevated transmit power level $P_{\text{tx_boosted}}$. The SNR is then recalculated and devices that now meet or exceed $\text{SNR}_{\text{threshold}}$ are included in the set of candidates $I_{\text{candidate}}$ for scheduling and optimization.

C. Scheduling Phase

After SNR estimation and forming the candidate device set $I_{\text{candidate}}$, the resource allocation phase begins, as detailed in Algorithm 1. There are two stages in Algorithm 1, stage 1 is for the URLLC device and stage 2 is for mMTC device. Resources are distributed according to the priority and QoS categories, prioritizing URLLC services for low-latency needs. Thus, URLLC devices receive resource allocation first.

Initially, we prioritize URLLC devices in $\text{DeviceList}_{\text{URLLC}}$ meeting $\text{SNR}_{\text{threshold}}$, sorting them by latency requirements L_i in ascending order to address latency-sensitive devices first. Each URLLC device is then classified into one of m QoS

Algorithm 1 Scheduling Phase - Joint URLLC/mMTC Resource Allocation

Input: $I_{\text{candidate}}$, $\text{DeviceList}_{\text{URLLC}}$, $\text{DeviceList}_{\text{mMTC}}$, QoS classes, initial transmit power $P_{\text{tx_global}}$, $\text{SNR}_{\text{threshold}}$

Output: scheduled devices, RB utilization

Stage 1: URLLC Device Scheduling (Priority = 1)

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1: Sort  $\text{DeviceList}_{\text{URLLC}}$  by increasing  $L_i$ 
2: for each device  $i \in \text{DeviceList}_{\text{URLLC}}$  do
3:   Get QoS class  $m \leftarrow \text{QoS\_Class}[i]$ 
4:   for each numerology  $\mu \in \text{MappingTable}[m]$  do
5:     if  $\text{RB}_{\text{Available}}[\mu] \geq \text{RB}_i$  then
6:       Allocate  $\text{RB}_i$  in numerology  $\mu$  to device  $i$ 
7:        $\text{RB}_{\text{Available}}[\mu] \leftarrow \text{RB}_{\text{Available}}[\mu] - \text{RB}_i$ 
8:        $\text{RB}_{\text{Used}}[\mu] \leftarrow \text{RB}_{\text{Used}}[\mu] + \text{RB}_i$ 
9:       Mark  $i$  as allocated
10:    break
11:   end if
12: end for
13: end for
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classes based on latency, mapped to a preferred numerology μ based on a mapping table. During resource allocation, the base station prioritizes these numerologies accordingly (Lines 2-4). If RBs remain available in all numerologies, the base station allocates one RB to the device i and updates the available resources. If the device i is marked as ‘allocated’, the current scheduling loop is completed (Lines 5-10). This phase ensures that URLLC devices are prioritized in bandwidth-constrained scenarios, demonstrating the algorithm’s delay-aware design. The subsequent phase involves assigning mMTC devices in stage 2 using the available numerology resources remaining from the first stage. The scheduling process is structurally identical to that of URLLC, following the same QoS-based numerology mapping. To avoid redundancy, the detailed steps are omitted.

D. Computational Complexity Analysis

As shown in Algorithm 1, the proposed algorithm achieves $\mathcal{O}(I \log I)$ complexity, where I is the total number of devices, which makes it suitable for real-time deployment. Initialization, SNR estimation/boosting, and scheduling run in $\mathcal{O}(I)$, $\mathcal{O}(I) + \mathcal{O}(I \cdot \text{NLoS}_{\text{URLLC}})$, and $\mathcal{O}(I \cdot n_{\text{max}}) \approx \mathcal{O}(I)$, respectively, while priority-based sorting adds $\mathcal{O}(I \log I)$. Overall, this efficiency enables sub-millisecond resource allocation for URLLC services in industrial scenarios with hundreds of devices.

V. SIMULATION RESULTS

In this section, we present the simulation results of the Joint URLLC/mMTC resource allocation algorithm with the

optimization objective described in Section III.

A. Simulation Setup

The simulations were carried out using a MATLAB-based system-level model of a $50\text{m} \times 50\text{m}$ industrial factory. A base station 35m was placed outside the factory center at a height of 5m, with devices on the ground. The system operated at 3.5 GHz with 100 MHz bandwidth and 30 dBm transmit power. The noise figure of the device was set to 9 dB, while the antenna gains were configured as -15dBi for the base station and 3dBi for the devices. A total of 200 devices were randomly placed using Monte Carlo sampling, comprising 70% URLLC and 30% mMTC to reflect industrial service requirements. Three numerology options were supported: 15 kHz ($\mu = 0$), 30 kHz ($\mu = 1$), and 60 kHz ($\mu = 2$). URLLC employed 2-symbol mini-slots for low latency, while mMTC used 14-symbol normal-slots for higher efficiency.

B. Result and Discussion

Fig. 2 illustrates the SNR (dB) distribution of devices before and after enhancement in the initial Monte Carlo simulation, sorted by LoS and NLoS channels. Initially, some NLoS URLLC devices have SNRs below 20 dB (black dashed line), as shown in Fig. 2 (a), indicating reliability issues. To resolve this, we introduce a priority-based SNR-boosting strategy. NLoS URLLC devices with SNR below the threshold receive an 8 dB transmit power boost ($P_{\text{tx,boosted}} = P_{\text{tx}} + 8$), consistent with medium BS power limits [11].

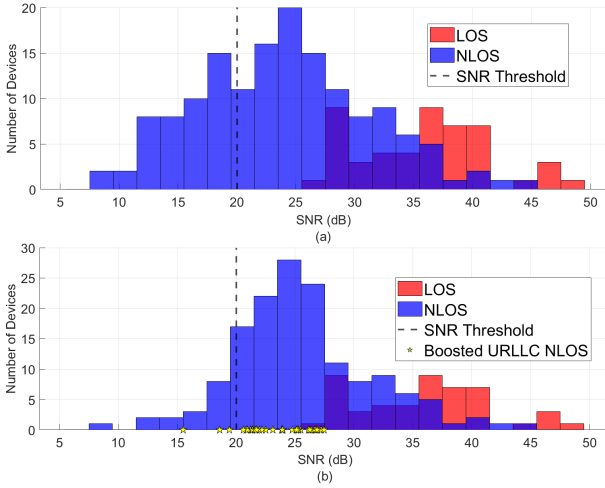


Fig. 2. SNR Distribution before and after Boosting. (a) Before Boosting, and (b) After Boosting.

Enhanced SNRs are recalculated considering PL, shadow fading, antenna gain, and noise in line with 6G requirements. From Fig. 2 (b), the proposed boost strategy raises a significant number of NLoS URLLC devices (yellow stars) above the SNR threshold, enabling resource allocation. This proactive boosting improves reliability and addresses fading in edge devices by improving crucial channel qualities without adding system complexity.

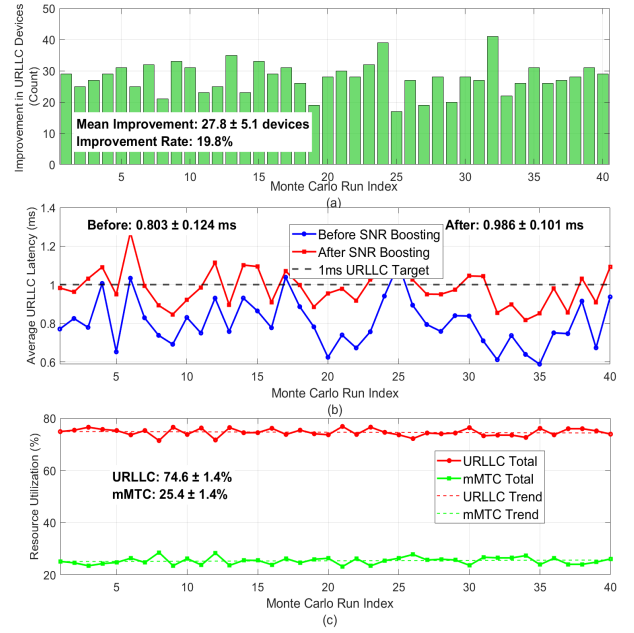


Fig. 3. Enhanced Monte Carlo Statistical Analysis Results. (a) SNR Boosting Improvement per Monte Carlo Run, (b) URLLC Latency Evolution Across 40 Monte Carlo Runs, and (c) Service Type Resource Utilization Comparison with Trends.

In wireless systems, stochastic elements like device location, PL, and shadow fading necessitate comprehensive evaluation beyond single-run simulations. To evaluate robustness, 40 separate Monte Carlo simulations were performed using different device distributions and PL conditions. Fig. 3 (a) shows SNR boosting added an average of 27.8 URLLC devices per run with ± 5.1 standard deviation, reflecting a 19.8% improvement in devices meeting SNR threshold criteria, confirming effectiveness under various stochastic conditions. Fig. 3 (b) shows average URLLC latency increased from $0.803 \pm 0.124\text{ ms}$ to $0.986 \pm 0.101\text{ ms}$ post-boosting, remaining well within the 1 ms requirement and satisfying Industrial IoT QoS demands; this increase results from longer queuing delays as a significantly larger device pool becomes schedulable post-boosting. Fig. 3 (c) shows URLLC occupied $74.6\% \pm 1.4\%$ of system resources while mMTC used $25.4\% \pm 1.4\%$, consistent with the 70% URLLC ratio and the prioritization strategy, with low fluctuation highlighting stable and differentiated resource allocation.

We evaluated the proposed algorithm using two key metrics: URLLC latency Cumulative Distribution Function (CDF) and packet loss rate. Since no existing benchmark jointly considers URLLC and mMTC coexistence under shared resource conditions, these classical schedulers were adopted as representative baselines for comparative evaluation. For a fair comparison under identical conditions where SNR boosting was applied to all algorithms, the proposed algorithm was compared against three baselines: Round Robin (RR) for fairness, Proportional Fair (PF) for fairness-throughput balance, and Earliest Deadline First (EDF) for latency prioritization. As shown in Fig. 4,

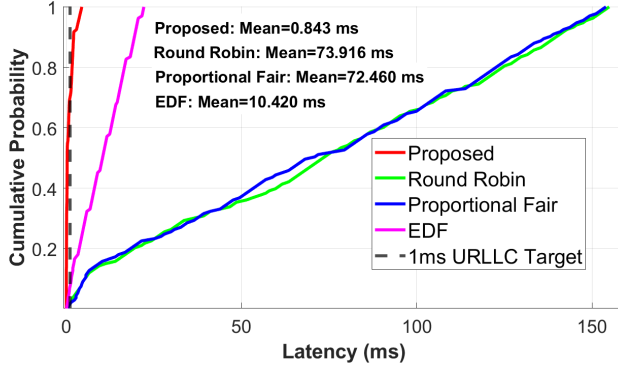


Fig. 4. Latency CDF Comparison.

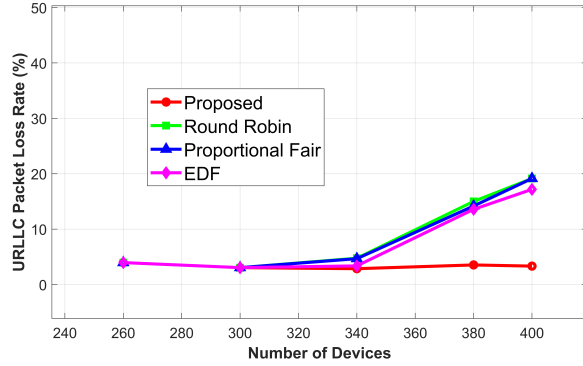


Fig. 5. Packet loss rate Comparison.

the proposed algorithm achieves 0.843 ms average URLLC latency, significantly outperforming EDF (10.420 ms), RR (73.916 ms), and PF (72.460 ms) through QoS-driven numerology mapping and latency-aware joint scheduling, while RR and PF overlook strict URLLC constraints and EDF misallocates numerologies despite being latency-oriented. Fig. 5 demonstrates robustness under heavy load, where the proposed algorithm maintains an approximate 3.2% packet loss rate as devices increase from 260 to 400 due to URLLC-prioritized scheduling, while RR, PF, and EDF experience steep increases toward 20%, confirming superior reliability and scalability under congestion.

This simulation results demonstrate the effectiveness of the proposed algorithm in meeting stringent URLLC QoS demands. However, the algorithm's strict URLLC priority guarantee means it is not throughput-optimal, exhibiting a performance trade-off compared to purely opportunistic schedulers such as PF. Despite this trade-off, the algorithm's advantages in critical performance metrics make it highly suitable for dynamic, multi-service industrial IoT scenarios.

VI. CONCLUSION AND FUTURE WORK

This paper demonstrates that by integrating flexible numerology with a priority-based SNR enhancement strategy,

a robust resource allocation framework is achieved for 6G smart factory environments. The simulation results confirm that the proposed framework increases the number of URLLC devices that meet scheduling requirements by 19.8%, while maintaining an average latency of 0.803 ms. Furthermore, the comparative analysis demonstrates that the proposed algorithm substantially outperforms conventional benchmarks, simultaneously reducing URLLC latency to sub-millisecond levels and ensuring high reliability under network congestion conditions. Future work will focus on applying reinforcement learning for adaptive optimization and extending the framework to multi-node cooperative scheduling to enhance scalability and system intelligence.

REFERENCES

- [1] A. Hazra, A. Munusamy, M. Adhikari, L. K. Awasthi *et al.*, "6g-enabled ultra-reliable low latency communication for industry 5.0: Challenges and future directions," *IEEE Communications Standards Magazine*, vol. 8, no. 2, pp. 36–42, 2024.
- [2] S. R. Pokhrel, J. Ding, J. Park, O.-S. Park, and J. Choi, "Towards enabling critical mmTc: A review of urllc within mmTc," *IEEE access*, vol. 8, pp. 131 796–131 813, 2020.
- [3] I. WP5D, "Future technology trends of terrestrial international mobile telecommunications systems towards 2030 and beyond," *International Telecommunication Union, Report M*, pp. 2516–0, 2022.
- [4] B. Adhikari, M. Jaseemuddin, and A. Anpalagan, "Resource allocation for co-existence of embb and urllc services in 6g wireless networks: A survey," *IEEE Access*, vol. 12, pp. 552–581, 2023.
- [5] J. Li, H. Wu, Y. Niu, B. Ai, N. Wang, and T. Q. Quek, "Resource allocation for downlink urllc in a smart factory," in *ICC 2024-IEEE International Conference on Communications*. IEEE, 2024, pp. 605–610.
- [6] M. Sharara, A. M. Masucci, Y. Carlinet, and N. Perrot, "Optimizing urllc resource allocation in open-ran: A transformer-based approach to flexible numerology," in *IEEE INFOCOM 2025-IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS)*, 2025.
- [7] Y. Sadi, S. Erkucuk, and E. Panayirci, "Flexible physical layer based resource allocation for machine type communications towards 6g," in *2020 2nd 6G Wireless Summit (6G SUMMIT)*. IEEE, 2020, pp. 1–5.
- [8] A. Yazar and H. Arslan, "Flexible multi-numerology systems for 5g new radio," *Journal of Mobile Multimedia*, vol. 14, no. 4, pp. 367–394, 2018.
- [9] L. Marijanovic, S. Schwarz, and M. Rupp, "Multi-user resource allocation for low latency communications based on mixed numerology," in *2019 IEEE 90th Vehicular Technology Conference (VTC2019-Fall)*. IEEE, 2019, pp. 1–7.
- [10] 3GPP, "NR; Base Station (BS) radio transmission and reception," 3rd Generation Partnership Project (3GPP), Technical Specification TS 38.104, 2024, v17.18.0, Release 17. [Online]. Available: https://www.3gpp.org/ftp/Specs/archive/38_series/38.104/38104-hv0.zip
- [11] 3GPP Technical Report, "Study on channel model for frequencies from 0.5 to 100 GHz," 3rd Generation Partnership Project (3GPP), Technical Report TR 38.901, Mar. 2022, release 17. [Online]. Available: https://www.3gpp.org/ftp/Specs/archive/38_series/38.901/
- [12] T. Jiang, J. Zhang, P. Tang, L. Tian, Y. Zheng, J. Dou, H. Asplund, L. Raschkowski, R. D'Errico, and T. Jämsä, "3GPP standardized 5G channel model for IIoT scenarios: A survey," *IEEE Internet of Things Journal*, vol. 8, no. 11, pp. 8799–8815, 2021.
- [13] J. Flores de Valgas, J. F. Monserrat, and H. Arslan, "Flexible numerology in 5G NR: Interference quantification and proper selection depending on the scenario," *Mobile Information Systems*, vol. 2021, no. 1, p. 6651326, 2021.
- [14] A. Yazar and H. Arslan, "Reliability enhancement in multi-numerology-based 5G new radio using ini-aware scheduling," *EURASIP Journal on Wireless Communications and Networking*, vol. 2019, no. 1, p. 110, 2019.